

上节回顾

谐振腔

$$1) k^2 = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 + \left(\frac{p\pi}{d}\right)^2 \quad \omega = kc \quad \lambda = \frac{2\pi}{k}$$

2) 对于 TE 波, $p \neq 0$, m 与 n 不能同时为 0.

对于 TM 波 p 可为 0, 但 m 与 n 均不为 0

$\left. \begin{array}{l} (m, n, p) \text{ 中最多有 } 1 \text{ 个为 } 0 \\ \text{最低阶模为 } (1, 0) \text{ 或 } (0, 1) \text{ 或 } (0, 1, 1) \end{array} \right\}$

电磁场的矢量与标势

$$\vec{B} = \nabla \times \vec{A}$$

$$\vec{E} = -\nabla\varphi - \frac{\partial \vec{A}}{\partial t}$$

$\vec{A}$ 与 φ 不唯一, 需加规范条件以限制

达朗贝尔方程 (洛伦兹规范 $\nabla \cdot \vec{A} + \frac{1}{c^2} \frac{\partial \varphi}{\partial t} = 0$)

$$\nabla^2 \vec{A} - \frac{1}{c^2} \frac{\partial^2 \vec{A}}{\partial t^2} = -\mu_0 \vec{j}$$

$$\nabla^2 \varphi - \frac{1}{c^2} \frac{\partial^2 \varphi}{\partial t^2} = -\rho/\epsilon_0$$

推迟势

求解思路 点电荷 + 叠加原理

$$\nabla^2 \varphi - \frac{1}{c^2} \frac{\partial^2 \varphi}{\partial t^2} = -(\rho u) \delta(\vec{r}) / \epsilon_0$$

在 $r \neq 0$ 处 球坐标系下

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \varphi}{\partial r} \right) - \frac{1}{c^2} \frac{\partial^2 \varphi}{\partial t^2} = 0$$

$$\varphi = \frac{u}{r}$$

$$\frac{\partial^2 u}{\partial r^2} - \frac{1}{c^2} \frac{\partial^2 u}{\partial t^2} = 0 \quad (\text{波动方程})$$

$$u(r, t) = f\left(t - \frac{r}{c}\right) + g\left(t + \frac{r}{c}\right)$$

$$\text{将 } u \text{ 代入 } \nabla^2 \varphi - \frac{1}{c^2} \frac{\partial^2 \varphi}{\partial t^2} = -\rho/\epsilon_0$$

$$f(t) = \frac{\rho u}{4\pi\epsilon_0}$$

$$\varphi(\vec{r}, t) = \frac{\rho(t - \frac{r}{c})}{4\pi\epsilon_0 r} = \frac{\rho(\vec{r}'=0, t - \frac{r}{c})}{4\pi\epsilon_0 r}$$

位于 $\vec{r}'$ 的点电荷激发的势

$$\varphi(\vec{r}, t) = \frac{\rho(\vec{r}', t - \frac{r}{c})}{4\pi\epsilon_0 r}$$

--- 任意电荷分布 ---

$$\varphi(\vec{r}, t) = \int \frac{\rho(\vec{r}', t - \frac{r}{c})}{4\pi\epsilon_0 r} dV'$$

$$\vec{A}(\vec{r}, t) = \mu_0 \int \frac{\vec{j}(\vec{r}', t - \frac{r}{c})}{4\pi r} dV'$$

辐射场的计算

$$\vec{j} \Rightarrow \vec{A} \Rightarrow \vec{B} = \nabla \times \vec{A}$$

$$\nabla \cdot \vec{j} = -\frac{\partial \rho}{\partial t} = i\omega \rho$$

$$\rho \Rightarrow \frac{\varphi}{A} \Rightarrow \vec{E}$$

在 $\vec{j}=0$ 的区域

$$\nabla \times \vec{B} = \frac{1}{c^2} \frac{\partial^2 \vec{E}}{\partial t^2} = -i\frac{\omega}{c^2} \vec{E} = -i\frac{k}{c} \vec{E}$$

$$\vec{E} = i\frac{c}{k} \nabla \times \vec{B}$$

电偶极子辐射

两个金属小球相距为 d ，用一个细导线相连。在 t 时刻，上面的小球带电 $q(t)$ ，下面的小球带电 $-q(t)$ 。其中 $q(t) = q_0 \cos \omega t$ 。这样产生一个振荡的电偶极子。求 $\vec{B}$ 与 $\vec{E}$ 。

解：推迟势为

$$\begin{aligned} \varphi(\vec{r}, t) &= \frac{1}{4\pi\epsilon_0} \left\{ \frac{q(t - \frac{r_+}{c})}{r_+} - \frac{q(t - \frac{r_-}{c})}{r_-} \right\} \\ &= \frac{1}{4\pi\epsilon_0} \left\{ \frac{q_0 \cos[\omega(t - \frac{r_+}{c})]}{r_+} - \frac{q_0 \cos[\omega(t - \frac{r_-}{c})]}{r_-} \right\} \end{aligned}$$

当 $d \ll R$ 时

$$r_{\pm} \approx R \mp \frac{d}{2} \cos \theta$$

$$\frac{1}{r_{\pm}} \approx \frac{1}{R \mp \frac{d}{2} \cos \theta} \approx \frac{1}{R} \left(1 \pm \frac{d}{2R} \cos \theta \right)$$

$$\cos[\omega(t - \frac{r_{\pm}}{c})] \approx \cos[\omega(t - \frac{R}{c}) \pm \frac{1}{2} k d \cos \theta]$$

$$= \cos[\omega(t - \frac{R}{c})] \cos(\frac{1}{2} k d \cos \theta) \mp \sin[\omega(t - \frac{R}{c})] \sin(\frac{1}{2} k d \cos \theta)$$

当 $k d \ll 1$ 即 $d \ll \lambda$ 时

$$\cos[\omega(t - \frac{r_{\pm}}{c})] \approx \cos[\omega(t - \frac{R}{c})] \mp \frac{1}{2} k d \cos \theta \sin[\omega(t - \frac{R}{c})]$$

$$\begin{aligned} \varphi(\vec{r}, t) &\approx \frac{q_0}{4\pi\epsilon_0 R} \left\{ -k d \cos \theta \sin[\omega(t - \frac{R}{c})] + \frac{d}{R} \cos \theta \cos[\omega(t - \frac{R}{c})] \right\} \\ &= \frac{P_0 \cos \theta}{4\pi\epsilon_0 R} \left\{ -k \sin[\omega(t - \frac{R}{c})] + \frac{1}{R} \cos[\omega(t - \frac{R}{c})] \right\} \end{aligned}$$

因此，

$$1) \omega = 0 \text{ 时 } \varphi(\vec{r}, t) = \frac{P_0 \cos \theta}{4\pi\epsilon_0 R^2} \quad (\text{静止电偶极子的势})$$

2) 在远场区 $R \gg \lambda$ ，第一项起主导作用

$$\varphi(\vec{r}, t) = - \frac{P_0 \omega}{4\pi\epsilon_0 c} \left(\frac{\cos \theta}{R} \right) \sin[\omega(t - \frac{R}{c})]$$

矢量 $\vec{A}$ 由 $\vec{E}$ 确定

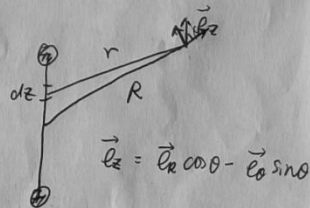
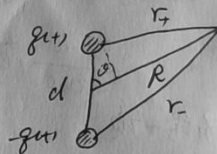
$$\vec{E}(t) = \frac{d\vec{q}}{dt} \vec{e}_z = -q_0 \omega \sin \omega t \vec{e}_z$$

$$\vec{A}(\vec{r}, t) = \frac{\mu_0}{4\pi} \int_{-\frac{d}{2}}^{\frac{d}{2}} \frac{-q_0 \omega \sin[\omega(t - \frac{r}{c})]}{r} dz \vec{e}_z$$

$$\approx - \frac{\mu_0 R \omega}{4\pi R} \sin[\omega(t - \frac{R}{c})] \vec{e}_z = \frac{\mu_0 \dot{P}(t - \frac{R}{c})}{4\pi R}$$

$$\vec{B} = \nabla \times \vec{A} = \frac{1}{r} \left[\frac{\partial}{\partial r} (r A_{\theta}) - \frac{\partial A_r}{\partial \theta} \right] \vec{e}_{\phi} = \frac{\mu_0 P_0 \omega^2}{4\pi c} \left(\frac{\sin \theta}{R} \right) \cos[\omega(t - \frac{R}{c})] \vec{e}_{\phi}$$

$$\vec{E} = -\nabla \varphi - \frac{\partial \vec{A}}{\partial t} = - \frac{\mu_0 P_0 \omega^2}{4\pi} \left(\frac{\sin \theta}{R} \right) \cos[\omega(t - \frac{R}{c})] \vec{e}_{\theta}$$



$$\vec{S} = \frac{1}{\mu_0} (\vec{E} \times \vec{B}) = \frac{\mu_0}{c} \left\{ \frac{P_0 \omega^2}{4\pi} \frac{\sin\theta}{R} \cos[\omega(t - r/c)] \right\}^2 \vec{e}_r$$

$$\langle \vec{S} \rangle = \left(\frac{\mu_0 P_0^2 \omega^4}{32\pi^2 c} \right) \frac{\sin^2\theta}{R^2} \vec{e}_r$$

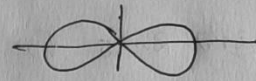
辐射功率

$$\langle P_{\text{rad}} \rangle = \int \langle \vec{S} \rangle \cdot d\vec{a} = \frac{\mu_0 P_0^2 \omega^4}{12\pi c}$$

$$\langle \vec{S} \rangle \propto \frac{1}{R^2} P_0^2 \omega^4$$

辐射功率的要求

$$\text{角分布 } \langle f(\theta, \phi) \rangle = \frac{\langle \vec{r} \cdot d\vec{a} \rangle}{d\Omega} = \langle \vec{S} \rangle R^2 \theta$$



磁偶极子辐射

考虑一细导线构成半径为 b 的圆环，它通有交变电流 $I(t) = I_0 \cos \omega t$ 。这就是一个振荡的磁偶极子辐射

$$\vec{m}(t) = \pi b^2 I(t) \vec{e}_z = m_0 \cos(\omega t) \vec{e}_z$$

$$m_0 = \pi b^2 I_0$$

$$\vec{A} = \frac{\mu_0}{4\pi} \int \frac{I_0 \cos[\omega(t - r/c)]}{r} d\vec{\ell}$$

对于一个在 x 轴上方任意点， $\vec{A}$ 指向 y 方向

$$\vec{A} = \frac{\mu_0 I_0 b}{4\pi} \vec{e}_y \int_0^{2\pi} \frac{\cos[\omega(t - r/c)]}{r} \cos\phi d\phi$$

由余弦定理得

$$r = \sqrt{R^2 + b^2 - 2Rb \cos\phi}$$

$$Rb \cos\phi = \vec{R} \cdot \vec{b} \quad \vec{R} = R \sin\theta \vec{e}_x + R \cos\theta \vec{e}_z \quad \vec{b} = b \cos\phi \vec{e}_x + b \sin\phi \vec{e}_y$$

$$\vec{R} \cdot \vec{b} = Rb \sin\theta \cos\phi$$

$$r = \sqrt{R^2 + b^2 - 2Rb \sin\theta \cos\phi}$$

当 $b \ll R$ 时

$$r \approx R \left(1 - \frac{1}{R} \sin\theta \cos\phi \right) = R - b \sin\theta \cos\phi$$

$$\frac{1}{r} \approx \frac{1}{R} \left(1 + \frac{1}{R} \sin\theta \cos\phi \right)$$

$$\cos[\omega(t - \frac{r}{c})] \approx \cos[\omega(t - \frac{R}{c}) + \frac{\omega b}{c} \sin\theta \cos\phi]$$

$$= \cos[\omega(t - \frac{R}{c})] \cos[\frac{\omega b}{c} \sin\theta \cos\phi] - \sin[\omega(t - \frac{R}{c})] \sin[\frac{\omega b}{c} \sin\theta \cos\phi]$$

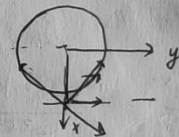
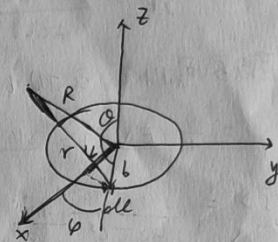
当 $k_b \ll 1$ 即 $b \ll \lambda$ 时 ($\frac{\omega b}{c} \ll 1$)

$$\cos[\omega(t - \frac{r}{c})] \approx \cos[\omega(t - \frac{R}{c})] - \sin[\omega(t - \frac{R}{c})] \frac{\omega b}{c} \sin\theta \cos\phi$$

$$\vec{A} \approx \frac{\mu_0 I_0 b}{4\pi R} \vec{e}_y \int_0^{2\pi} \left[\cos[\omega(t - \frac{R}{c})] - \sin[\omega(t - \frac{R}{c})] \frac{\omega b}{c} \sin\theta \cos\phi + \frac{b}{R} \sin\theta \cos\phi \cos[\omega(t - \frac{R}{c})] \right] \cos\phi d\phi$$

$$\int_0^{2\pi} \cos\phi d\phi = 0 \quad \int_0^{2\pi} \cos^2\phi d\phi = \int_0^{2\pi} \frac{1 + \cos 2\phi}{2} d\phi = \pi$$

$$\vec{A} \approx \frac{\mu_0 I_0 b^2}{4\pi R} \frac{\sin\theta}{R} \left\{ \frac{1}{R} \cos[\omega(t - \frac{R}{c})] - \frac{\omega}{c} \sin[\omega(t - \frac{R}{c})] \right\}$$



讨论

1) 静态下 ($\omega=0$)

$$\vec{A}(\vec{r}, 0) = \frac{\mu_0}{4\pi} \frac{m_0 \sin\theta}{R^2} \vec{e}_\phi$$

2) 在远场 $R \gg \frac{c}{\omega}$ 第二级近似作用

$$\vec{A}(\vec{r}, t) \approx -\frac{\mu_0 m_0 \omega}{4\pi c} \frac{\sin\theta}{R} \sin\left[\omega\left(t - \frac{R}{c}\right)\right] \vec{e}_\phi = i \frac{\mu_0 \omega}{4\pi R c} \vec{e}_\phi \times \vec{m}\left(t - \frac{R}{c}\right)$$

$$\vec{B} = \nabla \times \vec{A} = -\frac{\mu_0 m_0 \omega^2}{4\pi c^2} \left(\frac{\sin\theta}{R}\right) \cos\left[\omega\left(t - \frac{R}{c}\right)\right] \vec{e}_\theta$$

$$\vec{E} = -\frac{\partial \vec{A}}{\partial t} = \frac{\mu_0 m_0 \omega^2}{4\pi c} \frac{\sin\theta}{R} \sin\left[\omega\left(t - \frac{R}{c}\right)\right] \vec{e}_\theta$$

$$\vec{p} \rightarrow \frac{\vec{m}}{c}, \vec{E} \rightarrow c\vec{B}, c\vec{B} \rightarrow -\vec{E}$$

即自由电偶极子场给出磁偶极子

$\vec{E}$ 与 $\vec{B}$ 同相位, 互相垂直, 而且都垂直于传播方向($\vec{e}_r$)。振幅比 $E_0/B_0 = c$

$$\vec{S} = \frac{1}{\mu_0} (\vec{E} \times \vec{B}) = \frac{\mu_0}{c} \left\{ \frac{m_0 \omega^2}{4\pi c} \frac{\sin\theta}{R} \cos\left[\omega\left(t - \frac{R}{c}\right)\right] \right\}^2 \vec{e}_r$$

$$\langle \vec{S} \rangle = \frac{\mu_0 m_0^2 \omega^4}{32\pi^2 c^3} \frac{\sin^2\theta}{R^2} \vec{e}_r$$

$$\text{总辐射功率 } \langle P \rangle = \frac{\mu_0 m_0^2 \omega^4}{12\pi c^3}$$

电偶极子电流 $I_0 = q_0 \omega$, 设 $d = \tau b$

$$\frac{\langle P_{\text{电}} \rangle}{\langle P_{\text{磁}} \rangle} = \frac{\frac{\mu_0 m_0^2 \omega^4}{12\pi c^3} \cdot \frac{12\pi c}{\mu_0 p_0^2 \omega^4}}{\left(\frac{m_0}{p_0 c}\right)^2} = \left(\frac{\tau b^2 I_0}{q_0 d c}\right)^2 = \left(\frac{\tau b^2 I_0}{\frac{q_0}{\omega} \tau b c}\right)^2 = \left(\frac{\omega b}{c}\right)^2 \ll 1$$

电偶极子辐射的能量远大于磁偶极子